

SiguradoBaguio

Community Safety and Disaster Response Application

Document Version	1.1
Status	Pre-development — Planning Complete
Target Pilot	CDRRMO Baguio City
Pitch Target	Smart City Baguio Initiative
Platform	Flutter (Mobile), Next.js (Web)
Expected Launch Window	August to October 2026

1. Executive Overview

SiguradoBaguio is a purpose-built civic technology platform engineered for the residents, local government units (LGUs), and first responders of Baguio City. Situated in a Zone 3 seismic hazard area and along major typhoon corridors, Baguio City requires a highly resilient, always-available emergency communication infrastructure.

The application consolidates five primary civic functions: natural disaster alerting, relief goods coordination, accident reporting, civic complaints, and personal SOS signaling. Designed as both a thesis submission and a strategic pitch for the Smart City Baguio initiative, it is built for institutional adoption by the City Disaster Risk Reduction and Management Office (CDRRMO).

1.1 Problem Statement

- Baguio City endures 8 to 12 typhoons annually and is prone to landslides and flash flooding.
- Current disaster communication channels (radio, social media, SMS blasts) are fragmented and lack universal reach.
- There is no centralized, verifiable platform for citizens to report hazards, request rescue, or locate relief supplies.
- Cellular networks are highly susceptible to congestion or failure during active disasters, rendering standard applications useless.

1.2 The Solution

SiguradoBaguio resolves these critical vulnerabilities through three architectural commitments:

- **Offline-First Architecture:** The system degrades gracefully across four distinct connectivity tiers to ensure continuous operation without internet access.
- **Three-Tier Identity Model:** Every report and alert is cryptographically tied to a verified author, eliminating anonymity and false alarms.
- **Artificial Intelligence Integration:** AI handles initial triage and classification, significantly reducing the manual workload for LGU staff during high-volume disaster events.

1.3 Target User Groups

User Group	Verification Method	Primary Capabilities
Citizens	Philippine mobile number (OTP)	Receive alerts, submit reports, view relief centers, trigger SOS.
First Responders	Official credential/ID	Accept assignments, share live locations, mark incidents resolved.
LGU Officers	Government ID + TOTP	Publish disaster alerts, post relief centers, approve responders, access analytics.

2. Feature Specifications

2.1 Natural Disaster Alert System

This module enables verified LGU officers to broadcast hyper-localized emergency notifications.

- **Delivery Channels:** Simultaneous push notifications, SMS, and in-app banners.
- **Data Integration:** Surfaces official PAGASA and PHIVOLCS advisories directly into the LGU dashboard.
- **Operational Behaviors:** Alerts are geographically scoped to specific barangays, capped at 160 characters for SMS compatibility, and cached locally for 48 hours. Cell Broadcast SMS (WEA) is utilized via DICT partnership for device-agnostic delivery.
- **Strict Constraint:** AI models are explicitly prohibited from publishing alerts autonomously; human LGU officers must review and approve all broadcasts to prevent public panic.

2.2 Relief Goods Locator

A geographic tracking system for emergency supplies (food, water, medicine, shelter materials).

- **Management:** Only LGU officers and first responders can post supply centers.
- **Mapping:** Utilizes OpenStreetMap tiles with PostGIS ST_Distance sorting.
- **Lifecycle:** Centers auto-expire after 24 hours to prevent stale data; citizens can flag centers as "empty," triggering a re-verification ping.

2.3 Accident Alert & Crowdsourcing Verification

A citizen-driven reporting tool for active road hazards and vehicle accidents.

- **Verification Protocol:** An alert is only verified and pushed to responders if three independent user accounts report an incident within 500 meters of each other during a 10-minute window.
- **Anti-Abuse:** Submitting more than three reports within five minutes triggers an LGU manual review hold.

2.4 Report and Complaint System

A structured ticketing system for civic issues (e.g., road damage, flooding, illegal structures).

- **Routing:** Automatically directs formatted tickets to the correct LGU desk.
- **Offline Behavior:** Reports drafted offline are committed to a local SQLite database and synchronized to the server immediately upon reconnection.

2.5 SOS Emergency Signal

A rapid-deployment distress beacon built with multi-layered redundancy.

- **Activation:** Requires a three-second button hold to prevent accidental triggers.
- **Primary Delivery:** Notifies the five nearest verified first responders with GPS coordinates; auto-SMS is sent to emergency contacts.
- **Tier 2 Fallback (SMS Only):** Transmits pre-formatted SOS data via SMS to emergency contacts and 911.
- **Tier 1 Fallback (No Connectivity):** Broadcasts a Bluetooth Low Energy (BLE) beacon every two seconds. Nearby devices running SiguradoBaguio will detect the beacon and upload the distress signal on the victim's behalf once they regain connectivity.

3. System Architecture

3.1 Layered Infrastructure

- **Client Layer:** Flutter (Android/iOS), Next.js (Web Dashboard), SMS/WEA endpoints.
- **API Layer (Fastify Node.js):** Microservice-inspired logic handling Alerts, Auth, Reports, Synchronization, Geo-routing, and Notifications.
- **Data Layer:** PostgreSQL (PostGIS enabled), Redis (caching and rate-limiting), Supabase (object storage), Semaphore (SMS gateway).

3.2 Connectivity Tier Degradation Model

Tier	State	Capabilities	Technical Mechanism
4	Full Internet	Real-time sync, live maps, full features.	FCM, Supabase Realtime, REST API.
3	Intermittent	Background sync, optimistic UI updates.	SQLite, WorkManager/BGTaskScheduler.
2	SMS Only	Disaster alerts, SOS transmission.	Semaphore SMS, WEA Cell Broadcast.
1	None	Cached maps/alerts, BLE SOS beacon.	SQLite read, BLE broadcast, auto-dial.

4. Artificial Intelligence Integrations

AI features enhance operational efficiency but are strictly subordinate to manual fallbacks.

- **Report Photo Classifier (TFLite):** An EfficientNet-Lite model runs locally to suggest incident categories based on user-uploaded photos.
- **Duplicate Report Detection (Server):** Uses a multilingual sentence-transformer model and pgvector cosine similarity queries (>0.85 threshold) to automatically merge redundant reports submitted within 300 meters.
- **Risk Hotspot Prediction (XGBoost):** Analyzes historical incidents, MGB geohazards, elevation, and PAGASA rainfall data to generate a real-time risk choropleth map. This aligns methodologically with the AGRISCAN soil analysis project.

- **Offline Emergency Chatbot:** A quantized Phi-3 Mini SLM provides offline first-aid and evacuation guidance in English, Filipino, and Ilocano.
- **Credential OCR (Tesseract):** Extracts names and expiry dates from uploaded responder IDs to expedite LGU manual reviews.
- **Automated Analytics (Claude API):** Generates plain-language weekly situation reports from anonymized incident statistics for CDRRMO staff.

5. Development Timeline & Milestones

- **Phase 0: Foundation (Weeks 1-3):** Infrastructure scaffolding, DB schema migrations, Auth services, and offline-first synchronization implementation.
- **Phase 1: Core Features (Weeks 4-7):** Civic reporting modules, relief goods mapping, LGU web dashboard setup, and FCM notification pipelines.
- **Phase 2: Safety-Critical Features (Weeks 8-12):** LGU disaster alerts, crowdsourced accident verification, multi-tier SOS signals, and offline evacuation routing. *Gate: Requires written CDRRMO sign-off.*
- **Phase 3: AI Layer (Weeks 13-16):** Integration of TFLite classifiers, duplicate detection vectors, XGBoost hotspot mapping, and OCR pipelines.
- **Phase 4: Hardening & Launch (Weeks 17-20):** Penetration testing, load testing, closed CDRRMO beta in 3 barangays, app store submissions, and Smart City Baguio pitching.

6. Security and Compliance

- **Authentication:** SMS OTPs expire after 5 minutes and are strictly rate-limited. LGU portal utilizes TOTP with 30-minute session timeouts.
- **Data Privacy (NPC Compliant):** Phone numbers are bcrypt-hashed. Extracted responder credential images are immediately purged from object storage post-OCR.
- **API Security:** Full JWT authentication. Strict IP rate-limiting applied globally (60/min reads, 20/min writes) with HSTS and TLS 1.2+ enforcement.